

Project Grandmaster: An Autonomous Robotic Opponent for Go

PURPOSE

WHAT IS THE AIM?

This report aims to build an autonomous robot that can master the ancient game of Go.

WHAT ARE THE OBJECTIVES?

- Design and build a complete 3-DOF robotic arm electromechanical assembly capable of reaching all 25 board positions within the planar workspace.
- Design and build a Go-stone gripper capable of reliably picking up and placing stones.
- Implement inverse kinematics converting board coordinates to joint angles with $\approx \pm 2$ mm positional accuracy.
- Develop an interactive graphical user interface transmitting commands to the motion controller in under one second.

SPECIFICATIONS

links	lengths	joints	range	type	motor
Link 1	110mm	Joint 1	270°	revolute	DSS-M15S
Link 2	90mm	Joint 2	270°	revolute	DSS-M15S
Link 3	15mm	Joint 3	5mm	prismatic	DMS-MG90

Table 1: Robot Specifications

END-EFFECTOR

The end-effector combines Joint 3 and the gripping mechanism into a single unit. A servo converts angular motion into vertical movement via a horn and thin rod. At the bottom sits a permanent magnet for passive gripping of magnetic Go stones. It operates in three states:

- Low: picking position
- Hover: carrying/travelling position
- High: dropping position

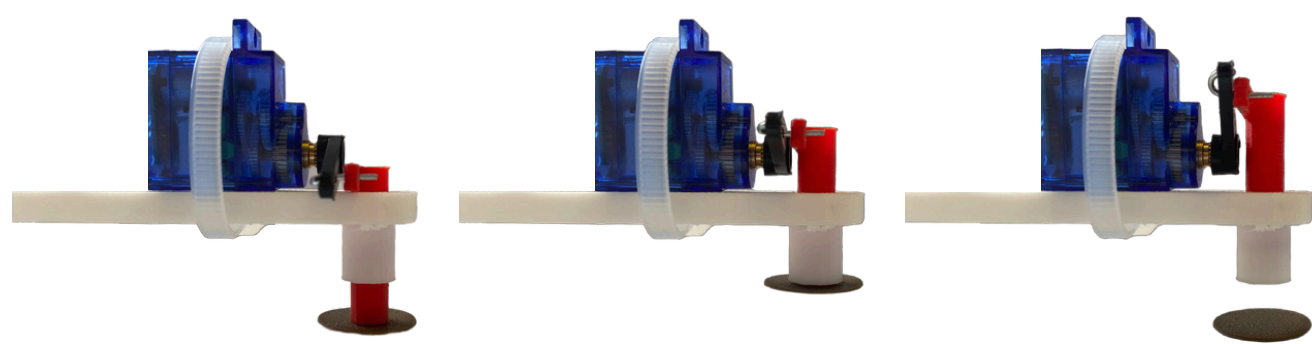


Figure 2: End-effector Mechanism

WORKFLOW

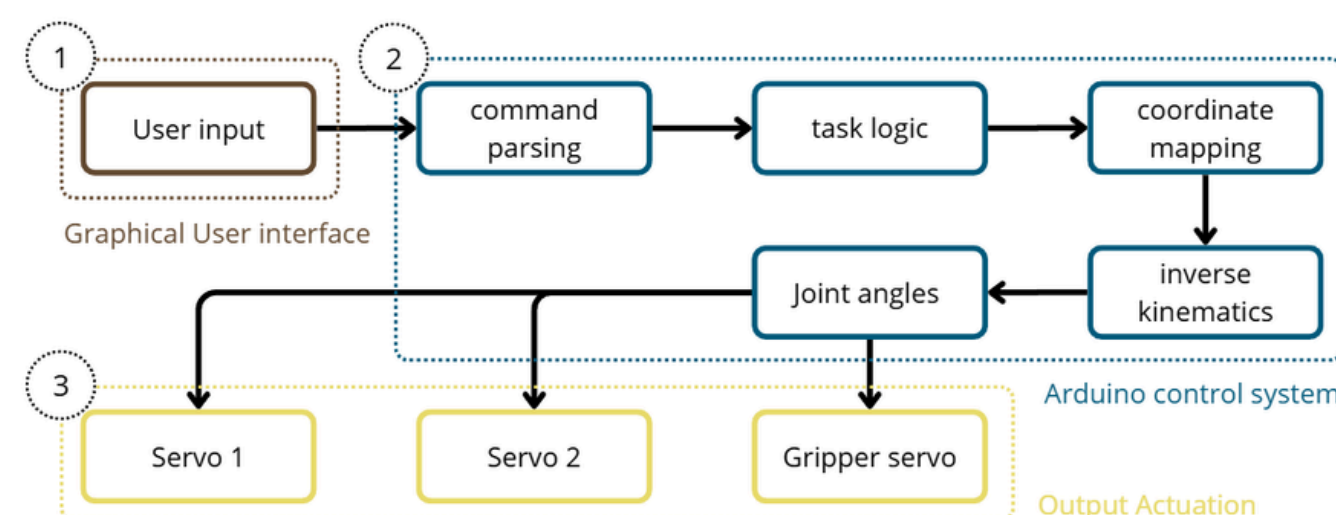


Figure 3: System Workflow

- User clicks a board intersection on the Python GUI
- GUI formats a command (e.g. B3P = column B, row 3, Place)
- Command sent to Arduino via USB serial communication
- Arduino parses command and determines action
- Target mapped to Cartesian coordinates
- Inverse kinematics computes joint angles
- PWM signals sent to servos, arm moves and executes task

MOTIVATION

WHY DO WE NEED A ROBOT?

Despite the resurgence of physical board games, a fundamental barrier persists: **you need another player**. Research shows physical **materiality** is a key part of the game experience that digital formats cannot replicate. Drawing upon those **challenges**, this report aims to build an autonomous robot that can master the ancient game of Go.

WHAT IS GO?

It is an ancient Chinese board game conceived more than 2,500 years ago. The goal is to capture more territory than your opponent. The players do so each in turn by placing stones on the intersections of the board. the board can vary in size but the game is commonly played on a 19x19 board. It has very simple rules and concepts, and yet it is considered by many people to be the world's greatest strategic game.

DEVICE

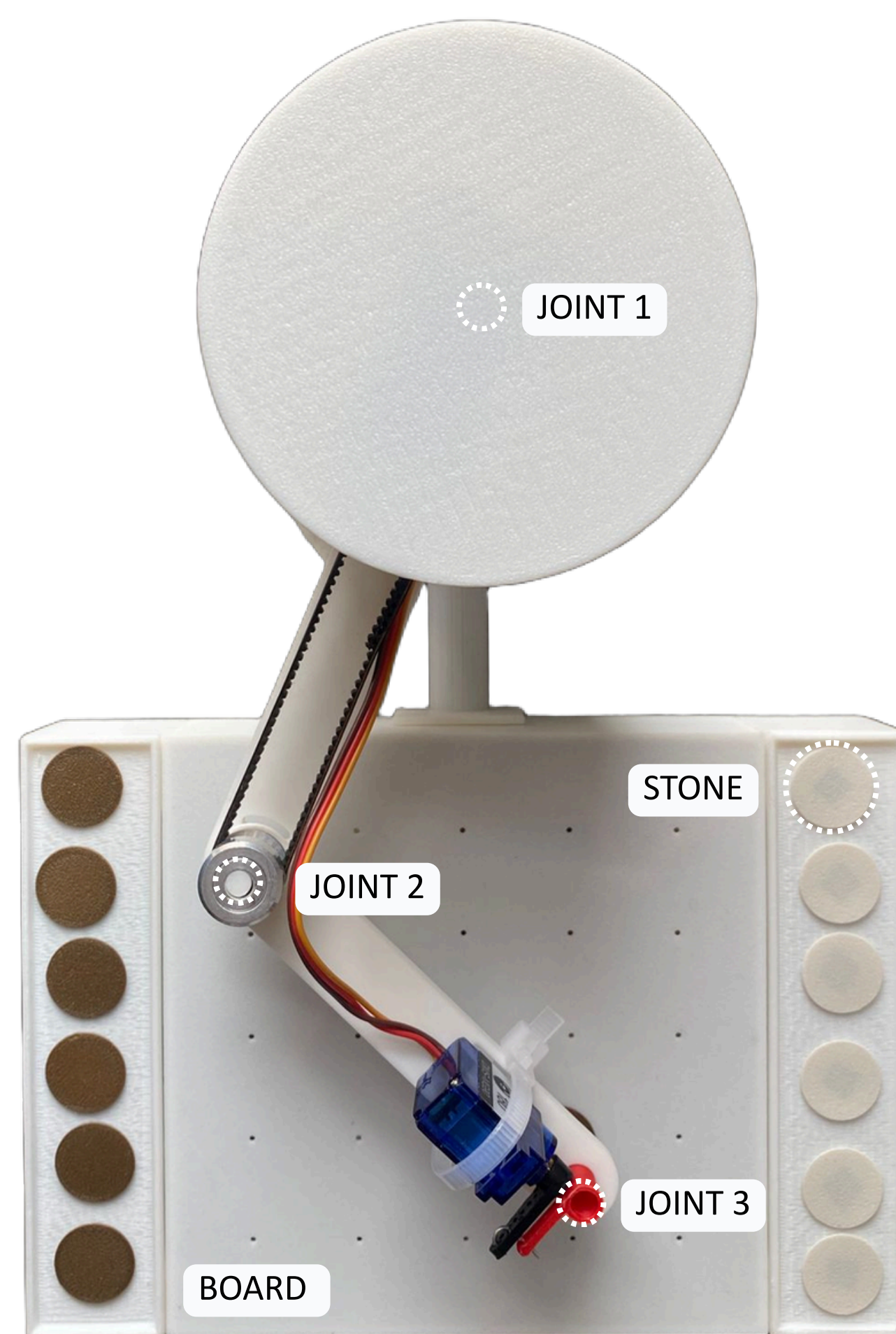


Figure 1: Go Playing Robot

RESULTS

- 100%** pick success rate achieved under controlled conditions (end-effector directly above stone)
- Positional accuracy varied due to backlash at Joint 2, the ± 2 mm target was not consistently achieved
- Adjustable motion **speed** via Arduino, set as slow
- All **25 board intersections** reached within defined workspace

LIMITATIONS

- PLA deformation $\rightarrow$ slack and **backlash** $\rightarrow$ low accuracy
- Open-loop** control $\rightarrow$ no feedback $\rightarrow$ no correction
- Limited workspace** $\rightarrow$ 2 unreachable positions
- Magnetic gripper $\rightarrow$ standard Go stones **incompatibility**
- limited vertical range $\rightarrow$ standard Go stones incompatibility

ARCHITECTURE

3-DOF SCARA robotic arm with RRP joint configuration. Controlled by an microcontroller, operated through a GUI. Joint 2 is operated remotely with a timing belt and pulley driving mechanism. Robot is at a fixed distance from the board.

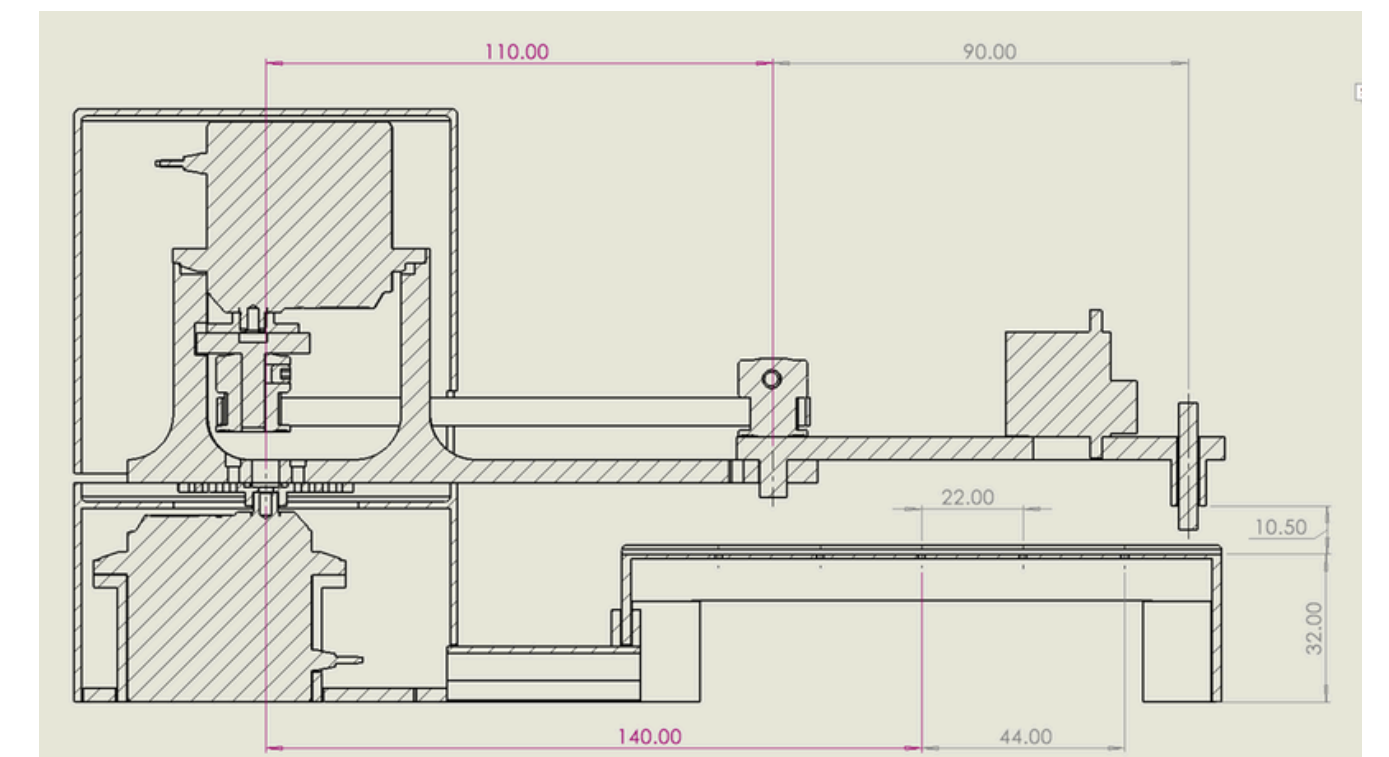


Figure 6: Slice of Assembly Drawing

CIRCUIT

Three servo motors are controlled by an Arduino Uno via PWM signals. Each servo is powered by an external 6V AA battery supply, with a common ground shared between the Arduino and power supply. Permanent connections are soldered to a stripboard to ensure reliability. The Arduino connects to the laptop via USB for serial communication with the GUI.

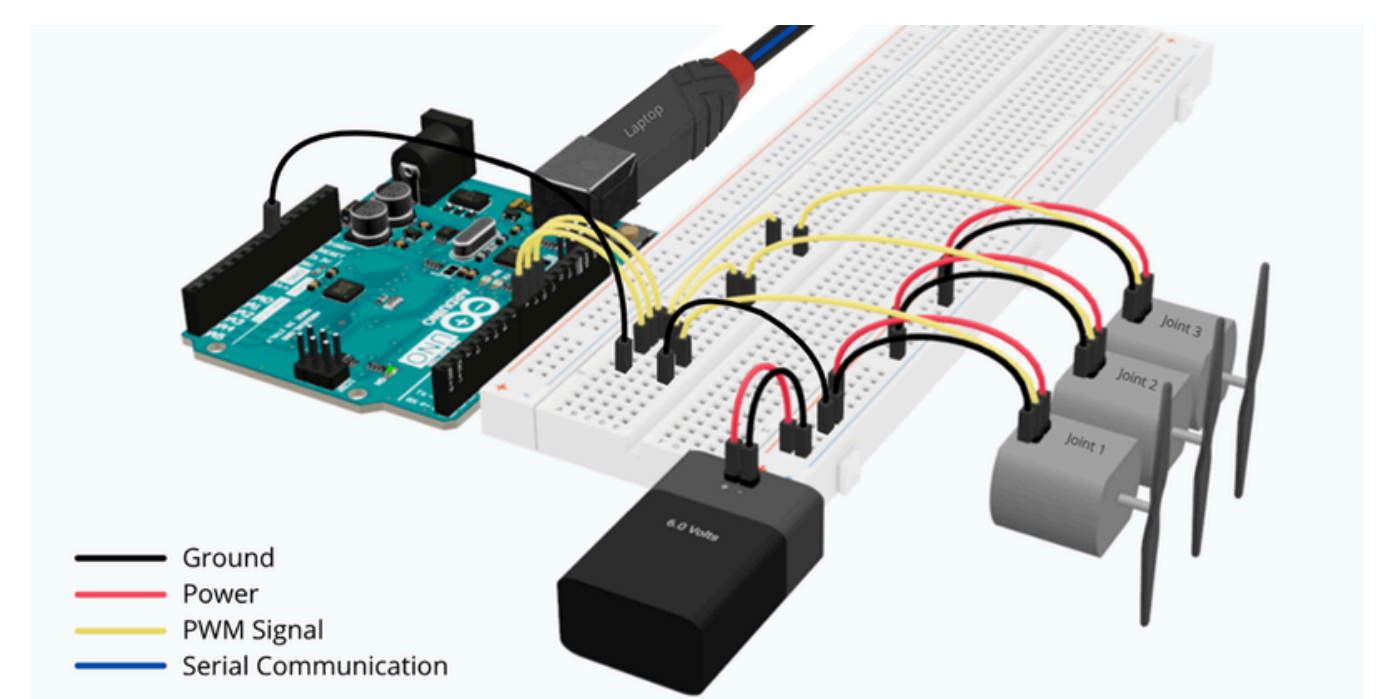


Figure 4: System Circuit Model

USER INTERFACE

A Python Tkinter graphical user interface was developed to control the robotic arm. The interface displays an interactive 5x5 Go board. The user places or removes stones by left or right clicking on board intersections. Each click generates a serial command (e.g. B3P) transmitted instantly to the Arduino.

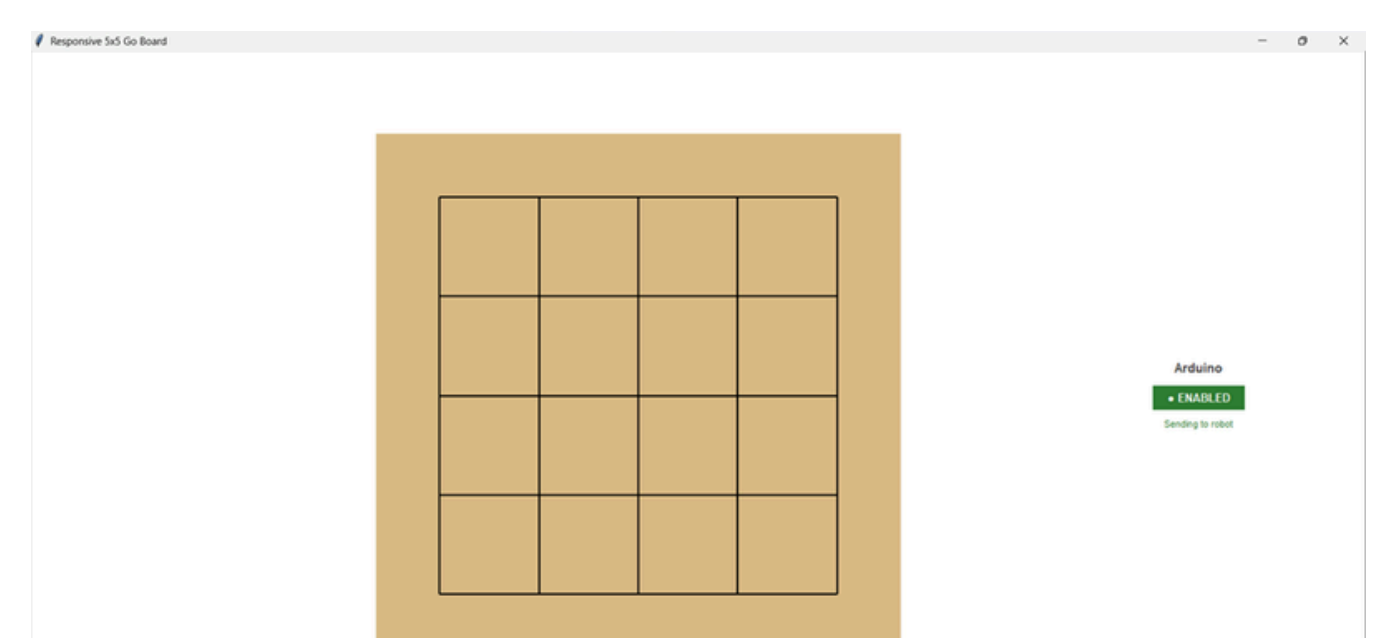


Figure 5: GUI Board Display

FUTURE WORK

- Computer vision** for real-time board state detection
- Go **AI engine** integration for full autonomy
- Closed-loop** feedback control to improve precision
- Redesigned end-effector for standard Go stones